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(2 + 1)-维色散长波系统的对称约化与群不变解

杜海清

(淮阴师范学院 数学系, 江苏 淮安, 223300)

摘 要: 研究了 (2 + 1)-维色散长波系统. 首先, 确定该系统的对称群, 此对称群含有 3 个任意的光滑函数, 然后利用某些子群, 把系统约化为热传导方程和第二型 Painlevé 方程.

关键词: (2 + 1)-维色散长波系统; 对称群; 群不变解

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1 (2 + 1)-维色散长波系统的对称群

本节研究 (2 + 1)-维色散长波系统

$$u_{yt} + v_{xx} + \frac{1}{2}(u^2)_{xy} = 0, \quad (1)$$

$$v_t + (uv + u + u_{xy})_x = 0,$$

的对称群. 该系统是由 Boiti 等人首先提出的, 它是由一个弱 Lax 对的相容性条件而得到的.

考虑系统式 (1) 的对称群

$$(u(x, t, y), v(x, t, y)) \rightarrow (\hat{u}(\hat{x}, \hat{t}, \hat{y}), \hat{v}(\hat{x}, \hat{t}, \hat{y})), \quad (2)$$

其中

$$\begin{aligned} \hat{x} &= x + \varepsilon \xi(x, t, y, u, v) + O(\varepsilon^2), \\ \hat{t} &= t + \varepsilon \tau(x, t, y, u, v) + O(\varepsilon^2), \\ \hat{y} &= y + \varepsilon \lambda(x, t, y, u, v) + O(\varepsilon^2), \\ \hat{u} &= u + \varepsilon \eta(x, t, y, u, v) + O(\varepsilon^2), \\ \hat{v} &= v + \varepsilon \chi(x, t, y, u, v) + O(\varepsilon^2), \end{aligned} \quad (3)$$

且 ε 是无穷小量, 而 $\xi, \tau, \lambda, \eta, \chi$ 都是光滑的待定函数.

式 (2) 的意思是: 如果 $u(x, t, y), v(x, t, y)$ 是式 (1) 的解, 那么 $\hat{u}(\hat{x}, \hat{t}, \hat{y}), \hat{v}(\hat{x}, \hat{t}, \hat{y})$ 也是式 (1) 的解.

设对称群式 (2) 所对应的无穷小生成元为

$$X = \xi \partial_x + \tau \partial_t + \lambda \partial_y + \eta \partial_u + \chi \partial_v, \quad (4)$$

无穷小生成元的延拓为

$$\text{Pr}(X) = X + \eta^x \partial_{u_x} + \eta^y \partial_{u_y} + \eta^{yt} \partial_{u_{yt}} + \eta^{xy} \partial_{u_{xy}} + \eta^{xxy} \partial_{u_{xxy}} + \chi^t \partial_{v_t} + \chi^x \partial_{v_x} + \chi^{xx} \partial_{v_{xx}}. \quad (5)$$

系数 $\eta^x, \eta^y, \eta^{yt}, \eta^{xy}, \eta^{xxy}, \chi^t, \chi^x, \chi^{xx}$ 由下面的式子确定

$$\begin{aligned} \eta^x &= D_x(\eta - \xi u_x - \tau u_t - \lambda u_y) + \xi u_{xx} + \tau u_{xt} + \lambda u_{xy}, \\ \eta^y &= D_y(\eta - \xi u_x - \tau u_t - \lambda u_y) + \xi u_{xy} + \tau u_{yt} + \lambda u_{yy}, \\ \eta^{yt} &= D_{yt}(\eta - \xi u_x - \tau u_t - \lambda u_y) + \xi u_{xyt} + \tau u_{yut} + \lambda u_{yyt}, \\ \eta^{xy} &= D_{xy}(\eta - \xi u_x - \tau u_t - \lambda u_y) + \xi u_{xxy} + \tau u_{xyt} + \lambda u_{xyy}, \end{aligned}$$

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作者简介: 杜海清 (1966-), 男, 江苏涟水人, 淮阴师范学院数学系讲师, 硕士.

$$\begin{aligned}
 \eta^{xy} &= D_{xy}(\eta - \xi u_x - \tau u_t - \lambda u_y) + \xi u_{xxy} + \tau u_{xyt} + \lambda u_{xyy}, \\
 \chi^t &= D_t(\eta - \xi v_x - \tau v_t - \lambda v_y) + \xi v_{xt} + \tau v_{tt} + \lambda v_{yt}, \\
 \chi^x &= D_x(\eta - \xi v_x - \tau v_t - \lambda v_y) + \xi v_{xx} + \tau v_{xt} + \lambda v_{xy}, \\
 \chi^{xx} &= D_{xx}(\eta - \xi v_x - \tau v_t - \lambda v_y) + \xi v_{xxx} + \tau v_{xxt} + \lambda v_{xxy}.
 \end{aligned} \quad (6)$$

另一方面, $\eta^x, \eta^y, \eta^{yt}, \eta^{xy}, \eta^{xxy}, \chi^t, \chi^x, \chi^{xx}$ 又满足下面的线性对称条件

$$\begin{aligned}
 \eta^{yt} &= -\chi^{xx} - \eta^x u_y - \eta^y u_x - \eta u_{xy} - \eta^{xy} u, \\
 \chi^t &= -\eta^x v - \chi u_x - \eta v_x - \chi^x u - \eta^x - \eta^{xy},
 \end{aligned} \quad (7)$$

其中 $u(x, t, y), v(x, t, y)$ 是系统式(1)的解.

将式(6)代入式(7)便得到一组关于 $\xi, \tau, \lambda, \eta, \chi$ 的决定方程. 求解这组方程得到 $\xi, \tau, \lambda, \eta, \chi$ 具有如下形式

$$\begin{aligned}
 \xi &= \frac{1}{2}A'(t)x + B(t), \tau = A(t), \lambda = C(y), \\
 \eta &= -\frac{1}{2}A'(t)u + \frac{1}{2}A''(t)x + B'(t), \\
 \chi &= -\frac{1}{2}(A'(t) + 2C'(y))(v + 1),
 \end{aligned} \quad (8)$$

其中 $A(t), B(t)$ 和 $C(y)$ 都是任意的光滑函数.

对于 2 个无穷小生成元 X 和 Y , 定义它们的括号积 $[X, Y] = XY - YX$, 则所有的无穷小生成元在括号积 $[,]$ 下构成一个李代数. 从式(8)可以看出, 系统式(1)的对称群所对应的李代数是无穷维的.

2 约化系统和群不变解

在式(8)中适当选取任意函数 $A(t), B(t)$ 或 $C(y)$ 则可以把式(1)约化成熟知的方程, 如热传导方程(见式(33)), 第二型的 Painlevé 方程(见式(41)), Riccati 方程, 或给出约化方程的精确解.

先来讨论一种简单的情形, 即

$$A(t) = 2t, B(t) = 0, C(y) = y.$$

此时式(8)成为

$$\xi = x, \tau = 2t, \lambda = y, \eta = -u, \chi = -2(v + 1). \quad (9)$$

相应的特征方程为

$$\frac{dx}{x} = \frac{dt}{2t} = \frac{dy}{y} = \frac{du}{-u} = \frac{dv}{-2(v+1)}. \quad (10)$$

因此, 可以选取 2 个新的自变量

$$p = t^{-\frac{1}{2}}x, q = t^{-\frac{1}{2}}y, \quad (11)$$

而 u, v 取下面的形式

$$u = t^{-\frac{1}{2}}f(p, q), v = t^{-1}g(p, q) - 1, \quad (12)$$

其中 f, g 都是关于变量 p 和 q 的光滑函数.

由式(12), 即有

$$\begin{aligned}
 u_x &= t^{-1}f_p, u_y = t^{-1}f_q, u_{xy} = t^{-\frac{3}{2}}f_{pq}, \\
 u_{yt} &= -t^{-2}f_q - \frac{1}{2}t^{-\frac{5}{2}}(xf_{pq} + yf_{qq}), u_{xxy} = t^{-2}f_{ppq}, \\
 v_t &= -t^{-2}g - \frac{1}{2}t^{-\frac{5}{2}}(xg_p + yg_q), v_x = t^{-\frac{3}{2}}g_p, v_{xx} = t^{-2}g_{pp}.
 \end{aligned} \quad (13)$$

把式(13)代入式(1)得到

$$\begin{aligned}
 f_q + \frac{1}{2}(pf_{pq} + qf_{qq}) &= g_{pp} + (ff_q)_p, \\
 g + \frac{1}{2}(pg_p + qg_q) &= (fg)_p + f_{ppq}.
 \end{aligned} \quad (14)$$

假设 f 和 g 满足如下关系

$$g = f_q, \quad (15)$$

则式(14)中的2个方程合为1个方程

$$f_q + \frac{1}{2}(pf_{pq} + qf_{qq}) = f_{ppq} + (ff_q)_p. \quad (16)$$

下面寻找式(16)形式的解

$$f(p, q) = F(\rho), \rho = p + q, \quad (17)$$

则式(16)变为

$$F' + \frac{1}{2}\rho F'' = F''' + (FF')'. \quad (18)$$

对式(18)积分2次,得到 Bernoulli 方程

$$F' = \frac{1}{2}\rho F - \frac{1}{2}F^2, \quad (19)$$

其解为

$$F = \frac{2\exp(\rho^2/4)}{\int \exp(\rho^2/4) d\rho + \tilde{c}}, \quad (20)$$

其中 $\tilde{c}$ 为积分常数.

再由式(12), (15), (17) 和 (20) 便可得到系统式(1)的群不变解

$$\begin{aligned} u &= t^{-\frac{1}{2}} \frac{2\exp\left(\frac{\rho^2}{4}\right)}{\int \exp\left(\frac{\rho^2}{4}\right) d\rho + \tilde{c}}, \\ v &= t^{-1} \left\{ \frac{\rho \exp\left(\frac{\rho^2}{4}\right)}{\int \exp\left(\frac{\rho^2}{4}\right) d\rho + \tilde{c}} - \frac{2\exp\left(\frac{\rho^2}{4}\right)}{\left[\int \exp\left(\frac{\rho^2}{4}\right) d\rho + \tilde{c}\right]^2} \right\} - 1, \end{aligned} \quad (21)$$

其中 $\rho = t^{-\frac{1}{2}}(x + y)$, $\tilde{c}$ 为积分常数.

可以利用 Mathematica 作出系统式(1)精确解的图像. 图1显示, 当 $\tilde{c} = 0$ 时, $t = 1$ (第1列)、4 (第2列)、8 (第3列) 时群不变解式(21)的形状.

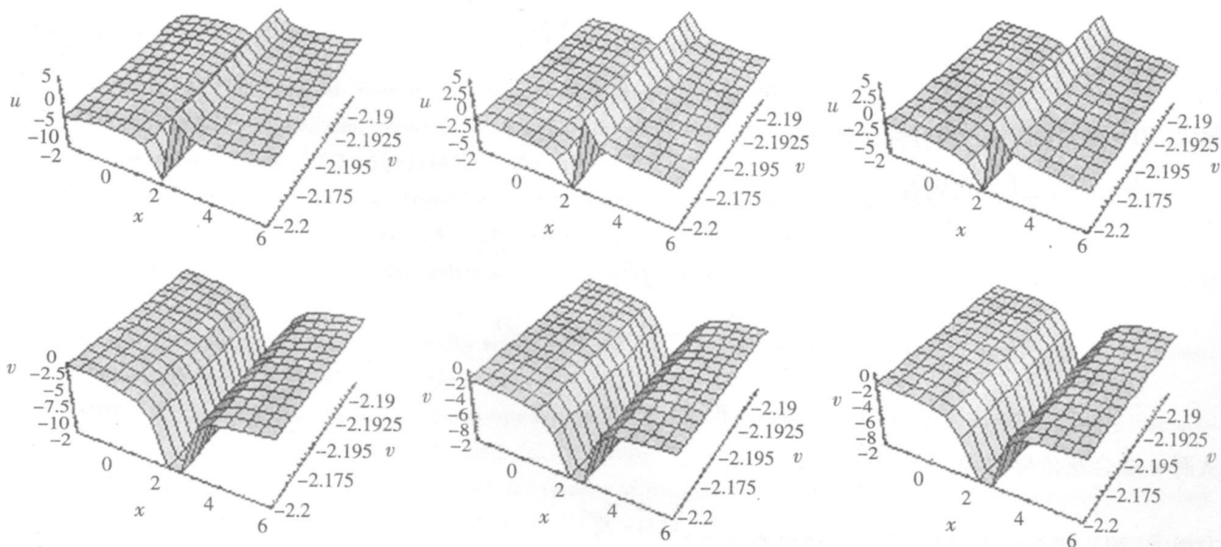


图1 群不变解的形状

下面再讨论如下情形: $A(t) \equiv 1$, $B(t)$ 和 $C(y)$ 是任意光滑函数.

此时,式(8) 成为

$$\xi = B(t), \tau = 1, \lambda = C(y), \eta = B'(t), \chi = -C'(y)(v+1). \quad (22)$$

相应的特征方程为

$$\frac{dx}{B(t)} = \frac{dt}{1} = \frac{dy}{C(y)} = \frac{du}{B'(t)} = \frac{dv}{-C'(y)(v+1)}. \quad (23)$$

因此,可以选取 2 个新的自变量

$$p = x - \int B(t) dt, q = \int \frac{dy}{C(y)} - t, \quad (24)$$

而 u, v 取下面的形式

$$u = B(t) + f(p, q), v = \frac{g(p, q)}{C(y)} - 1, \quad (25)$$

其中 f, g 都是关于变量 p 和 q 的光滑函数.

由式(25), 有

$$\begin{aligned} u_x &= f_p, u_y = \frac{f_q}{C(y)}, \\ u_{xy} &= \frac{f_{pq}}{C(y)}, u_{yt} = -\frac{B(t)f_{pq} + f_{qq}}{C(y)}, u_{xy} = \frac{f_{ppq}}{C(y)}, \\ v_t &= -\frac{B(t)g_p + g_q}{C(y)}, v_x = \frac{g_p}{C(y)}, v_{xx} = \frac{g_{pp}}{C(y)}. \end{aligned} \quad (26)$$

把式(26) 代入式(1) 得

$$\begin{aligned} f_{qq} &= g_{pp} + f_p f_q + f f_{pq}, \\ g_q &= (fg)_p + f_{ppq}. \end{aligned} \quad (27)$$

首先假设函数 f 和 g 满足如下关系

$$g = f_q, \quad (28)$$

则式(27) 中的 2 个方程合为 1 个方程

$$f_{qq} = f_{ppq} + f_p f_q + f f_{pq}. \quad (29)$$

式(29) 关于 q 积分 1 次, 得到

$$f_q = f_{pp} + f f_p. \quad (30)$$

引入势函数 ϕ , 得

$$\begin{aligned} \phi_q &= f_p + \frac{1}{2}f^2, \\ \phi_p &= f, \end{aligned} \quad (31)$$

则式(30) 是式(31) 的相容性条件.

再令 $w = \exp\left\{\frac{\phi}{2}\right\}$, 则有

$$\begin{aligned} w_p &= \frac{1}{2}wf, \\ w_{pp} &= \frac{1}{2}wf_p + \frac{1}{4}wf^2, \\ w_q &= \frac{1}{2}w\phi_q. \end{aligned} \quad (32)$$

从而 w 适合热传导方程

$$w_q = w_{pp}. \quad (33)$$

方程式(30) 和方程式(33) 之间的变换为

$$f = 2 \frac{w_p}{w}. \quad (34)$$

这样,由热传导方程式(33)的解 w ,就可以得到系统式(1)的解

$$\begin{aligned} u &= B(t) + 2 \frac{w_p}{w}, \\ v &= \frac{2}{C(y)} \left(\frac{w_{pq}}{w} - \frac{w_p w_q}{w^2} \right) - 1, \end{aligned} \quad (35)$$

其中 $p = x - \int B(t) dt$, $q = \int \frac{dy}{C(y)} - t$.

下面来寻找式(27)的行波解. 令

$$\rho = p + kq, f(p, q) = F(\rho), g(p, q) = G(\rho), \quad (36)$$

其中 k 是非零常数.

则式(27)变为

$$\begin{aligned} k^2 F'' &= G'' + kF'^2 + kFF'', \\ kG' &= (FG)' + kF'''. \end{aligned} \quad (37)$$

对式(37)的第1个方程积分2次,得到

$$G = k^2 F - \frac{k}{2} F^2 - c_1 \rho - c_2, \quad (38)$$

其中 c_1, c_2 是积分常数.

对式(37)的第2个方程积分1次,结合式(38)可得

$$F''' = \frac{1}{2} F^3 - \frac{3}{2} k F^2 + \left(k^2 + \frac{c_2}{k} \right) F + \frac{c_1}{k} \rho F - c_1 \rho - \left(c_2 + \frac{c_3}{k} \right), \quad (39)$$

其中 c_3 也是积分常数.

适当选取常数 k 以及积分常数 c_1, c_2, c_3 , 则方程式(39)可化成第二型 Painlevé 方程, 或者可以直接求解. 例如: $k = -\sqrt[3]{2}$, $c_1 = -\sqrt[3]{2}$, $c_2 = -1$, $c_3 = -\sqrt[3]{2}$. 则式(39)变为

$$F''' = \frac{1}{2} F^3 + \frac{3}{2} \sqrt[3]{2} F^2 + \left(\sqrt[3]{4} + \frac{1}{\sqrt[3]{2}} \right) F + \rho F + \sqrt[3]{2} \rho. \quad (40)$$

令 $K = \frac{1}{2}(F + \sqrt[3]{2})$, 则式(40)成为第二型 Painlevé 方程

$$K'' = 2K^3 + \rho K - \frac{1}{2}. \quad (41)$$

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$$\left| \sqrt{N} - \frac{L_n}{M_n} \right| = \left| \frac{M_n \sqrt{N} - L_n}{M_n} \right| = \frac{L_n^2 - NM_n^2}{M_n} \cdot \frac{1}{M_n \sqrt{N} + L_n} < \frac{1}{M_n} \cdot \frac{1}{M_n \sqrt{N} + M_n \sqrt{N}} = \frac{1}{2 \sqrt{N} M_n^2}.$$

例如

$$\sqrt{2} = \frac{3}{2} - \frac{1}{2^2 \cdot 3} - \frac{1}{2^3 \cdot 3 \cdot 17} - \frac{1}{2^4 \cdot 3 \cdot 17 \cdot 577} - \cdots,$$

若取前 4 项和作为 $\sqrt{2}$ 的近似值, 它的截断误差就小于

$$\frac{1}{2 \sqrt{2} (2^4 \cdot 3 \cdot 17 \cdot 577)^2} < 1.6 \times 10^{-12}.$$

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On Pell Series

HUANG Dong-ming

(College of Information Science and Technology, Hainan University, Haikou 570228, China)

Abstract: The purpose of this paper is to introduce the notion of Pell series. Some basic results on Pell series are established. As a main result, a particular Pell series extension of $\sqrt{N}$ is obtained.

Key words: Pell equation; Pell series; approximate calculation

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Symmetry Reducibility of (2 + 1) Dimensional Dispersive Long Wave System and the Group-invariant Solution

DU Hai-qing

(Department of Mathematics, Huaiyin Teachers' College, Huaian 223300, China)

Abstract: In this paper, we study on the (2 + 1) dimensional dispersive long-wave system. First, we determine the symmetry group of the system. It turns out that the symmetry group contains three arbitrary smooth functions. Then, we reduce the system to the heat equation and the second Painlevé equation by using some subgroups.

Key words: (2 + 1) dimensional dispersive long wave system; symmetry group; group invariant solution